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Sub Code: NEE 201

Paper Id: 120203

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B TECH
(SEM. II) THEORY EXAMINATION 2017-18
BASIC ELECTRICAL ENGINEERING

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. Assume missing data suitably, if any.

SECTION A

1. Attempt all questions in brief.

2 x 10 = 20

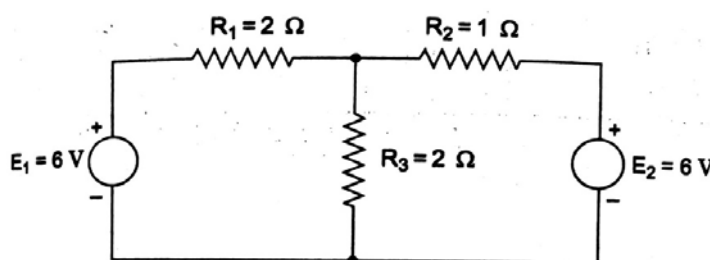
- a) What is maximum power transfer theorem?
- b) Distinguish between mesh & loop of network.
- c) What is the significance of peak factor?
- d) Why the inductors are usually iron cored?
- e) Why interconnection of 3-phase system is necessary?
- f) What are the essentials of indicating instruments?
- g) On what factors does the magnetic reluctance depend?
- h) What is a function of transformer?
- i) What is the function of armature in a DC generator?
- j) Why are induction motors called 'asynchronous'?

SECTION B

2. Attempt any three of the following:

10 x 3 = 30

- a) Find current in R_3 of the network by using Thevenin's theorem.



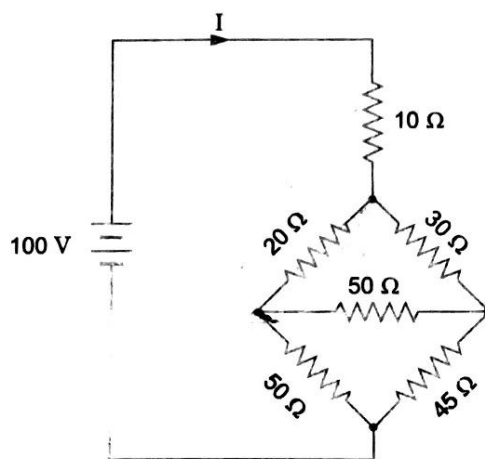
- b) Derive expressions for average value and rms value of a sinusoidally varying AC voltage.
- c) Explain two wattmeter method to measure 3-phase power in Y-connected load. Derive the phase angle in terms of wattmeter readings. Draw phasor diagram also.
- d) Discuss analogy between electric and magnetic circuits. On what factors does the magnetic reluctance depend? Is it constant at all flux densities?
- e) Enumerate all the parts of a DC machine. State the material and the function of each part.

SECTION C

3. Attempt any one parts of the following:

10 x 1 = 10

- What is the use of delta-star and star-delta transformation in the circuits? Explain the method of transforming a star network into a delta network.
- Find the current drawn I from the source voltage of 100 V in the circuit shown.



4. Attempt any one parts of the following:

10 x 1 = 10

- In a purely inductive circuit, prove from first principles, that the current lags behind applied voltage by quarter of a cycle and also show that the average power demand is zero.
- What are the causes of low power factor in supply system? Discuss its harming effects. How can the power factor be improved?

5. Attempt any one parts of the following:

10 x 1 = 10

- Derive the relationship between the line and phase voltages of an alternator.
- Derive the expression for torque produced in a moving coil type of instrument and explain, briefly, it's working.

6. Attempt any one parts of the following:

10 x 1 = 10

- Explain what are hysteresis and eddy current losses. Discuss how these losses are reduced in transformer cores?
- Draw and explain phasor diagram of transformer under loaded condition.

7. Attempt any one parts of the following:

10 x 1 = 10

- Explain the principle of operation of a 3-phase synchronous motor.
- Explain the principle of operation of a 1-phase induction motor.